

[page 1121, ex 76]

$$\nabla F = \frac{\partial}{\partial x}(e^x \sin y) + \frac{\partial}{\partial y}(e^y \sin z) + \frac{\partial}{\partial z}(e^z \sin x) = e^x \sin y + e^y \sin z + e^z \sin x$$

[page 1121, ex 13]

$$\text{curl } F = \nabla \times F = \begin{vmatrix} \frac{d}{dx} & \frac{d}{dy} & \frac{d}{dz} \\ y^2 z^3 & 2xy z^3 & 3xy^2 z^2 \end{vmatrix} =$$

$$= (6xyz^2 - 6xyz^2)\mathbf{i} - (3y^2 z^2 - 3y^2 z^2)\mathbf{j} + (2yz^3 - 2yz^3)\mathbf{k} = 0$$

By Theorem 4, F is conservative

$$F = \nabla f \quad f(x, y, z) = y^2 z^3$$

$$f(x, y, z) = x y^2 z^3 + g(y, z)$$

$$f_y(x, y, z) = 2xy z^3 + g_y(y, z)$$

$$f_y(x, y, z) = 2xy z^3 \Rightarrow g(y, z) = h(z)$$

$$f(x, y, z) = x y^2 z^3 + h(z)$$

$$f_z(x, y, z) = 3xy^2 z^2 + h'(z)$$

$$f_z(x, y, z) = 3xy^2z^2$$

$$h(z) = k$$

$$f(x, y, z) = xy^2z^3 + k$$

~~question 20~~

question 21

$$r(t) = (2 \cos t, 2 \sin t) \quad t \in [0, 2\pi]$$

$$dr = (-2 \sin t \, dt, 2 \cos t \, dt)$$

$$\oint_C F \, dr = \iint \text{curl}(F) \, dA$$

$$\text{curl}(F) = (2F_2/dx - 2F_1/dy)k =$$

$$= (2xy - (-2xy))k \quad r \in [0, 2]$$

3)3

$$\theta \in [0, 2\pi]$$

$$\iint_R \text{curl}(F) \, dA = \int_0^{2\pi} \int_0^2 4r^3 \cos \theta \, dr \, d\theta$$

$$= \int_0^{2\pi} \sin \theta \, d\theta$$

$$\iint_R \text{curl}(F) \, dA = 0$$

$$\oint_C F \, dr = 0$$

~~question 22~~

question 22)

$$\int_C \frac{-y}{x^2+y^2} dx + \frac{x}{x^2+y^2} dy$$

a) $x^2 + y^2 = 1$

b) $x^2 + \frac{y^2}{4} = 1$

~~we~~ we can use Green's Theorem

$$F = \left(\frac{-y}{x^2+y^2} \right) \vec{i} + \left(\frac{x}{x^2+y^2} \right) \vec{j}$$

$$\text{curl}(F) = \frac{dF_1}{dx} - \frac{dF_2}{dy}$$

$$= \left(\frac{1}{x^2+y^2} \right) \left(\frac{d \left(\frac{x}{x^2+y^2} \right)}{dy} - \frac{d \left(\frac{-y}{x^2+y^2} \right)}{dx} \right) =$$
$$= \frac{1}{x^2+y^2} \circ \left(\frac{2xy}{x^2+y^2} - \frac{2xy}{x^2+y^2} \right) = 0$$

so F is conservative vector

a) $\int_C F dz = \int_R \text{curl}(F) dA = 0$

b) $\int_C F dz = \int_R \text{curl}(F) dA = 0 = \text{line integral}$
 F is conservative